

When does the default PING circuit start to sing?

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Abstract

The default `snn.components.ping()` circuit is silent at weak drive, then switches abruptly into a strongly rhythmic state near 50 Hz per input channel. The dominant rhythm rises from 14.93 to 27.96 Hz across the active sweep and remains predominantly below the conventional gamma band. One compiled graph produces the entire response curve. No model parameter is tuned between conditions.

Methods

1. **Author one graph.** A typed 128-channel spike input drives the default PING component, which contains 80 excitatory (E) and 20 inhibitory (I) cells. The timestep is 0.1 ms. Neuron, synapse, delay, initializer, and constraint specifications remain at their component defaults.
2. **Vary only input drive.** Homogeneous Poisson input spans 0 Hz, 25 Hz, 50 Hz, 75 Hz, 100 Hz, 125 Hz, 150 Hz, 200 Hz per channel. Every condition uses the same compiled graph and 5 paired deterministic trials, each lasting 1000 ms. Measurements exclude the first 200 ms.
3. **Resolve the dominant rhythm.** The `default-ping-dominant-rhythm-v1` policy computes a single-trial Welch spectrum from the E-population raster. It searches 5–80 Hz, interpolates the peak between bins, rejects maxima at the band edge, and requires peak power to exceed 3 times the band median. If a peak at half the selected frequency carries at least 30% of its power, the lower frequency is reported. This prevents a strong harmonic from masquerading as the population rhythm.
4. **Score rhythmicity.** The standard `exp054` metric bins the E-population spike count at 1 ms, computes its normalized autocorrelogram, and applies a three-point smoothing kernel. If L is the first side-lobe height and T the following trough height, the dimensionless Michelson contrast is $R = \frac{L-T}{L+T}$. Silence is reported as $R = 0$.
5. **Measure E/I timing separately.** Post-transient population spikes are binned at 1 ms. The lag of the strongest E/I cross-correlation within plus or minus 20 ms is descriptive and does not enter the gamma-resolution rule.

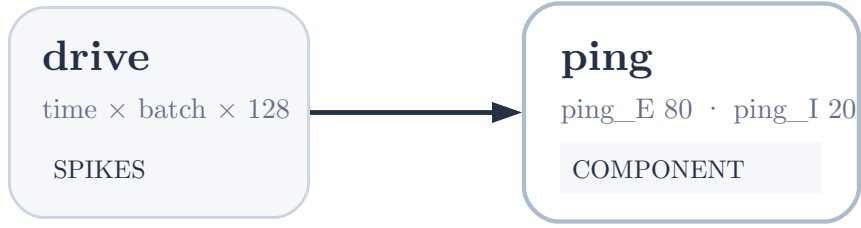


Figure 1: Compiled graph reused at every input condition. A time-by-batch-by-128 spike tensor drives one default PING component with 80 E cells and 20 I cells. Only the input event probability changes across the sweep; the graph and its parameters remain fixed.

Results

The network is silent through 25 Hz per channel. At 50 Hz it activates, the median rhythmicity score reaches 1, and the dominant rhythm is 14.93 Hz. Rhythmicity therefore appears discontinuously, while the rhythm frequency increases smoothly with drive. The default circuit sings, but under this protocol it mostly sings below gamma.

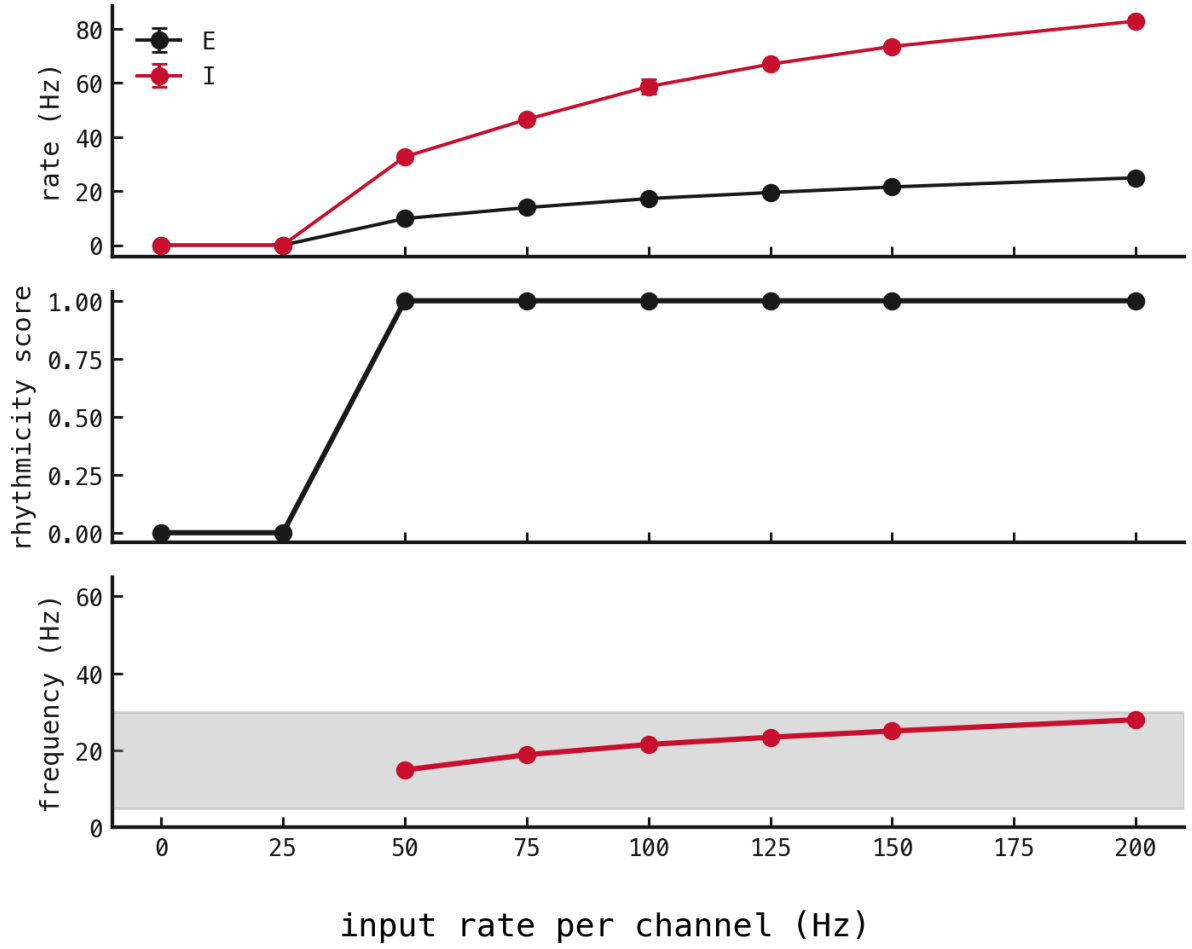


Figure 2: Response of one fixed default PING graph to homogeneous Poisson drive. Top: mean E and I firing rates in hertz, with standard deviations across 5 paired trials. Middle: median lobe-trough rhythmicity contrast R , with error bars showing half the interquartile range; $R = 0$ is flat and R approaches 1 as periodic structure strengthens. Bottom: median dominant E-population frequency in hertz after subharmonic correction; the grey region marks 5–30 Hz. Rhythmicity switches on abruptly at 50 Hz per channel, while the dominant rhythm increases smoothly from 14.93 Hz.

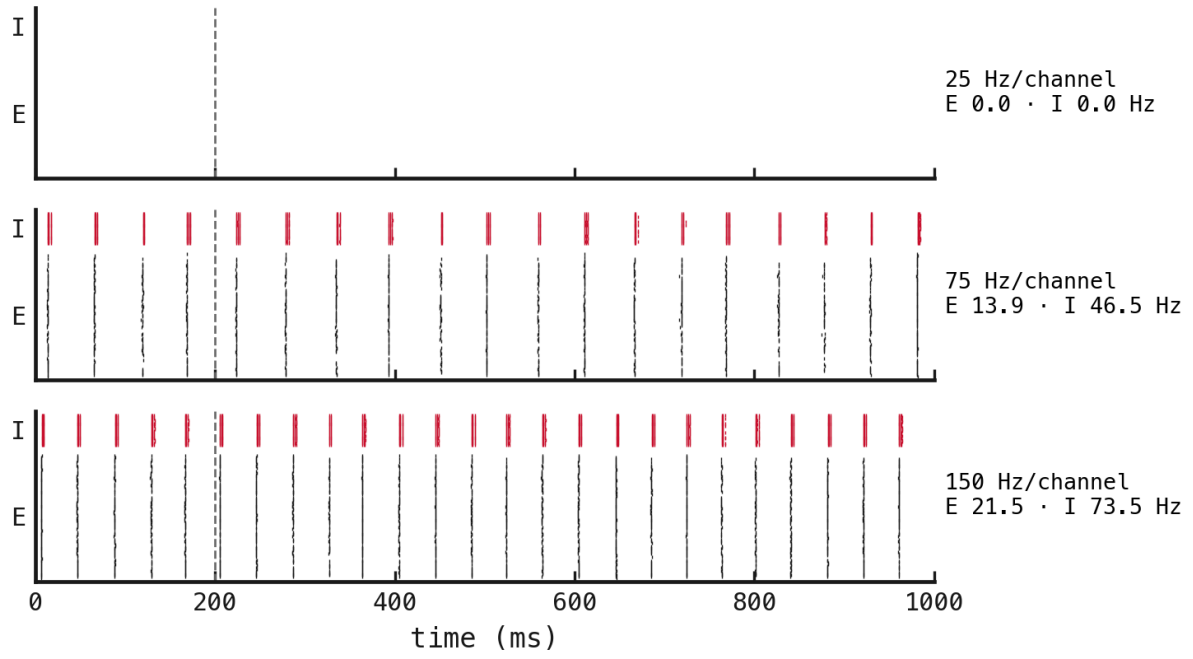


Figure 3: Single-trial E and I rasters at the three input rates fixed before the primary sweep. Time is in milliseconds; E cells are black, I cells are red, and the dashed vertical line marks the 200 ms transient exclusion. The right labels report input rate per channel and post-transient mean population rates in hertz. The 25 Hz condition is silent, whereas the higher-drive conditions show repeated population volleys.

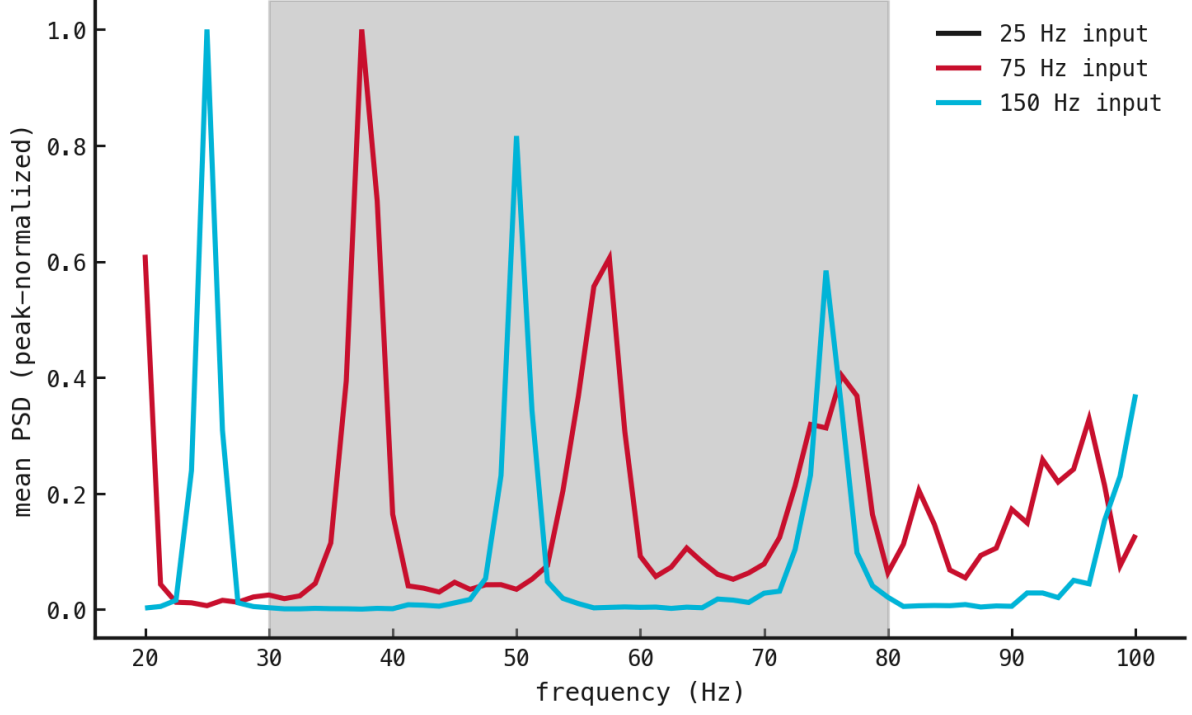


Figure 4: Mean E-population power spectra at the preselected input rates. Frequency is in hertz and power is normalized to each curve’s maximum for shape comparison; frequency resolution uses the unnormalized single-trial spectra. The grey region marks the former 30–80 Hz gamma-only window. The dominant peaks sit below it, while visible harmonics fall inside it, explaining why a gamma-restricted search gave a misleading frequency curve.